One Health Microbiome Center Co-laboratory

Background

Mission statement: The OHMC-CL exists to catalyze cutting edge microbiome science at PSU through facilitating independent access to cutting edge high-throughput technologies for microbiome profiling.

What the OHMC-CL is: The OHMC-CL is collaborative space with shared equipment, resources, and protocols for microbiome characterization. The goal is for users across PSU to be able to access and use these resources to enable their research.

What the OHMC-CL is not: The OHMC-CL is not a pay-for-service core. Typical core services are not provided. The ultimate success of any given project is the responsibility of the trainee or user. It is expected that all users will participate in scheduled training sessions and become independent users. Users who do not anticipate becoming independent users are instead encouraged to explore core services through the Huck Genomics Core or external service providers. Recommendations specific to the project can be provided upon request.

What has the OHMC-CL enabled? In the first year and a half of operation, users have profiled 6,140 samples from a variety of sources including human, mouse, soil, plant, livestock, poultry, and environmental sources. There have been a total of 34 users, from 22 labs, from 12 different departments. Users have performed a mixture of sample extraction, qPCR, amplicon, and metagenomic sequencing.

Why does the OHMC-CL exist?

Community support: There are 550 OHMC members across 42 departments, and 10 colleges at PSU. While our research interests are highly varied, many of us use the same ‘omics-based approaches to interrogate our microbial communities. The OHMC-CL exists to facilitate the sharing of technical knowledge and resources to lower boundaries to using ‘omics based approaches to study microbial communities.

Equipment: The equipment required to process microbiome samples for sequencing costs hundreds of thousands of dollars. When each lab purchases their own, redundancy and waste are inevitable with capital equipment sitting unused for most of its useful lifetime. Through a variety of sources, including a generous donation from Qiagen, the OHMC-CL enables new labs, or those who are getting into the microbiome space, to use shared equipment creating significant cost savings which may be reinvested into doing science to enable publications and grants.

Economy of scale/cost sharing: The major non-personnel costs of microbiome sequencing are derived from (i) extraction, (ii) library preparation, and (iii) sequencing. The OHMC-CL enables cost savings in each of these axes. Extraction: through allocations of extraction reagents provided by Qiagen, we can partially, or completely, reduce the major costs of extraction to only pipette tips and minor plastics disposables. Library preparation: through reaction scaling and custom protocols, the

cost of library prep can be reduced 5-10x. Further, by trainees making their own libraries, significant savings are derived from labor while also providing marketable and transferable skills. Sequencing: sequencing costs are highly dependent on throughput ranging from $3/Gb to $500/Gb depending on sequencer and chemistry used. In effect, it is possible to obtain amplicon sequencing at a cost of ~$4/sample (~100,000 reads/sample via NextSeq 2000 XLEAP reagents), and metagenomic sequencing at ~$30/sample (10Gb/sample via NovaSeq X 25B reagents). To reach these costs, hundreds of samples need to be sequenced at a time; however, most users do not have the required number of samples to reach this economy of scale. The OHMC-CL provides an opportunity to pool samples across labs to create cost savings for our community.

Becoming a OHMC-CL User

Intake Form and Meeting: All new users must complete an intake form available here using their PSU login. The information collected in this form is used to help report usage both internally and externally. It is requested that each new project performed in the OHMC-CL has its own intake form. After reviewing the form, we will reach out to schedule a meeting with the trainee and PI to discuss project needs and to identify appropriate equipment and protocols.

Responsibilities of Trainees:

1) Attend a training session before using any OHMC-CL equipment.
2) After training, operate equipment independently and responsibly.
3) Schedule equipment in advance and cancel reservations in a timely manner if needed.
4) Bring all necessary items and reagents with you to perform your work. Clean and reset equipment/workspaces after use.
5) Troubleshoot experiments with your PI. Ask for help via the OHMC-Co-lab Slack only after making a good-faith effort to resolve issues independently.
6) Contribute back to the co-lab by helping to train others, suggesting and testing protocol improvements, making amendments to protocols on github, and sharing tips via Slack.

Responsibilities of Supervisors:

1) Support trainees in troubleshooting and data interpretation within their home lab.
2) Reinforce expectations for independent use and respectful engagement with co-lab infrastructure.
3) Assist in evaluating whether a trainee's project is appropriate for the co-lab versus a fee-for-service option.

Communicating with other OHMC-CL Users: The preferred mode of communication for the OHMC-CL users is through a slack workgroup. An invite link is available here. Please send all communications through this channel including requests for help troubleshooting. This will facilitate disseminating knowledge across the user base and allowing users to help other users.

Training: Researchers will be trained to operate laboratory equipment used in microbiome workflows, including automated liquid handlers for 96-well nucleic acid extraction, library preparation, and qPCR. Training covers how to set up, troubleshoot, and maintain instruments. Sessions are held every other

month to support consistent onboarding and skill-building. The training schedule will be posted in the Training channel of the OHMC-CL slack group. Training is currently offered on:

1. High throughput DNA extraction using QIAcube HT and Powersoil reagents
2. Integra mini-96 pipettes
3. Primary PCR for amplicon sequencing
4. qPCR via the QIAquant 384
5. Tapestation analysis of DNA/RNA samples
6. Metagenomic library preparation

Access: Users will be able to access the OHMC-CL during normal business hours; however, card access will not be given due to biosafety protocols.

Requesting donated reagents: We receive an annual donation which is allocated among users. Best efforts are made to ensure allocations are evenly spread among lab groups; however, users with funded projects are encouraged to use their support to purchase consumables, thus allowing donated resources to seed future funding.

Scheduling equipment: Equipment is available on a first-come first-served basis and must be reserved via the link in each equipment's slack channel. Frequent last minute cancellation will lead to a loss of scheduling privileges.

Contributing to the OHMC-CL: Since the OHMC-CL runs on donated equipment, reagents, and time, it is expected that all users will give back to the OHMC-CL. Examples of how users may contribute include:

1. Becoming a trainer: users who have become competent with specific equipment will be invited to carry out training of new users.
2. Contributing to protocols: users are expected to contribute to both new and existing protocols. Protocols are available via github and may be modified freely. This could include clarifying wording, adding pictures, etc. Anything to help future users. Please make a pull request for changes to be merged into the protocol.

Shared Resources

Tissuelyzer III (Sample disruption/bead beater)

QIAcube HT (automated 96-well extraction)

Tapestation 4200 (automated electrophoresis for QC of nucleic acids)

QIAquant 384-well qPCR system

Integra mini-96 pipettes

-20°C co-lab freezer (limited, temporary storage)

Extraction hood (biosafety cabinet for nucleic acid work)

Illumina iSeq (pilot scale sequencing runs, metagenomic library QC and normalization)

Protocol repository (GitHub link): continually updated by users; includes validated and in-progress protocols for DNA extraction, library prep, and sequencing.

Unique Dual Indexes: Illumina DNA prep compatible indexes based on nextera design. 574 indexes are available which are compatible with most illumina PCR-based library preps or for amplicon sequencing.

Slack Community: user forum for troubleshooting, sharing knowledge, and training opportunities.

Sequencing through the OHMC-CL

For users wishing to include their samples for sequencing with other OHMC users, the following input specifications are required.

Amplicon: Only samples which have gone through primary PCR to generate an amplicon will be included for communal sequencing runs. These may be generated using any primer set and enzyme so long as the product contains the partial Nextera overlap for indexing. A recommended protocol is available here. We recommend using qPCR to generate amplicons; however, users requiring >30 cycles of PCR will need to use gel electrophoresis or other methods to ensure amplicons have formed. Users who have performed >30 cycles of PCR will be required to do a bead cleanup to remove excessive primer dimers that may cause issues downstream. Users will need to provide their samples in a skirted 96 well plate and include both a printed and digital copy of the layout using this template.

Metagenomic Sequencing: Users wishing to pool samples for metagenomic sequencing **must** request indexes before preparing their libraries. Indexes must be coordinated across users on the run to prevent loss of data. Metagenomic libraries should be at least 1nM in concentration with at least 20 µL. Use of a tapestation or equivalent is required to ensure an appropriate fragment size. Libraries may be directly submitted in skirted 96 well plates including a printed and digital copy of the layout using this template.

Cost Sharing for Sequencing: Users will be required to provide an IO. This IO will be used to purchase sequencing cartridges (amplicon) or to resolve P card charges (metagenomic sequencing). Actual costs will depend on filling the sequencing run. It may take 1-2 months to resolve charges.

User Acknowledgements

I, _____ (name), a _____ (current position), working in the lab of _____ (PI's name), have read this document in its entirety. I understand that the OHMC-Co-lab is a shared, collaborative space that requires personal responsibility and active participation. I agree to follow all co-lab expectations, operate equipment independently after training, and seek troubleshooting support within my home lab. I acknowledge that continued access is dependent on adherence to these guidelines.

Signature: _____

PI's Signature: _____

Date: _____